

SODIUM

Normal: 135-145 mEq/L

FUNCTION - WHAT NORMAL LOOKS LIKE

- Major extracellular electrolyte -> controls water balance, blood volume, and BP.
- Supports nerve impulses / neurologic function; regulated in part by aldosterone.

HYPONATREMIA

LOW: <135

Critical cutoff not stated in lecture

WHAT CAUSES IT TO DECREASE?

- GI / other loss: vomiting, NG suction, sweating
- Diuretics or renal sodium loss
- Water overload: CHF or hypotonic fluids
- SIADH -> too much ADH -> dilution
- Low intake / NPO

RECOGNIZE IT - SIGNS / SYMPTOMS

- Confusion, lethargy, trouble concentrating
- Seizures / stupor = severe neurologic danger
- Weakness; shallow respirations can be a late sign
- Abdominal cramps, decreased bowel sounds
- Low BP and decreased urine output

NURSING MANAGEMENT - WHAT DO I DO?

- Trend Na; frequent neuro checks, VS, respirations, I&O
- Seizure + fall precautions for neurologic changes
- If dilutional / SIADH: restrict fluids; diuretic as ordered
- Hypertonic IV as ordered; use close monitoring - fluid overload risk
- Treat cause; moderate sodium intake if appropriate

HYPERNATREMIA

HIGH: >145

Critical cutoff not stated in lecture

WHAT CAUSES IT TO INCREASE?

- Water loss: fever, sweating, vomiting, diarrhea
- Diabetes insipidus -> excess free-water loss
- Excess sodium: oral / IV, tube feeds, hypertonic fluid
- Corticosteroids or excess aldosterone -> retain sodium

RECOGNIZE IT - SIGNS / SYMPTOMS

- Thirst; dry mouth / skin
- Agitation or confusion
- Flushed skin and fever
- Fluid retention / edema when sodium gain is the cause
- Decreased urine output

NURSING MANAGEMENT - WHAT DO I DO?

- Trend Na; frequent neuro checks, VS, I&O, daily weight
- Safety / fall precautions for confusion
- Replace fluid with ordered isotonic or hypotonic IV solution
- Give slowly -> rapid correction risks cerebral edema
- Restrict sodium and treat cause (example: DI)

WHY / DANGER Low Na / dilution -> water enters cells -> brain swelling. High Na -> water leaves cells -> cellular dehydration.

DIET LOW: salty foods in moderation if appropriate. HIGH: limit canned foods, deli meat, frozen dinners, chips, pickles.

BETTER? Na trends to 135-145; mentation, BP, urine output, and hydration improve without seizures or fluid overload.

POTASSIUM

Normal: 3.5-5.0 mmol/L

FUNCTION - WHAT NORMAL LOOKS LIKE

- Main intracellular electrolyte -> supports nerve impulses and skeletal / cardiac muscle contraction.
- Heart rhythm is the priority; kidneys are the main regulator of potassium balance.

HYPOKALEMIA

LOW: <3.5

Lecture urgency cue: <2.5

HYPERKALEMIA

HIGH: >5.0

Critical cutoff not stated in lecture

WHAT CAUSES IT TO DECREASE?

- GI loss: vomiting, diarrhea, NG drainage
- Loop / thiazide diuretics (example: furosemide)
- Dehydration or poor intake
- Insulin or albuterol -> shifts K into cells
- Hyperaldosteronism / corticosteroids

WHAT CAUSES IT TO INCREASE?

- Renal impairment -> decreased K excretion
- K-retaining meds: ACE inhibitor, spironolactone, NSAID
- Addison disease -> too little aldosterone
- Burns / trauma -> K leaves injured cells
- Excess IV potassium

RECOGNIZE IT - SIGNS / SYMPTOMS

- Weak / irregular pulse; low BP or orthostasis
- Muscle weakness, cramps, decreased DTR; paralysis if severe
- Shallow respirations / diminished breath sounds
- Decreased bowel sounds / slowed GI motility
- ECG: ST depression, inverted T wave, prominent U wave

RECOGNIZE IT - SIGNS / SYMPTOMS

- Irregular heartbeat; bradycardia / low BP; VF if severe
- ECG: peaked T waves, ST elevation
- Muscle weakness / cramps; tingling, burning, numbness
- Decreased DTR; respiratory failure if severe
- Hyperactive bowel sounds / diarrhea; confusion

NURSING MANAGEMENT - WHAT DO I DO?

- Telemetry / ECG; monitor VS, respirations, GI, DTR
- Check K, Mg, renal function, and urine output
- Oral K with food when possible
- IV K: dilute, use pump, infuse slowly - NEVER IV push
- Monitor IV site; question K-wasting diuretic; watch digoxin toxicity

NURSING MANAGEMENT - WHAT DO I DO?

- Continuous ECG; assess VS, weakness, and respirations
- Stop K sources; question K-retaining meds; notify provider
- Protect heart: IV calcium gluconate as ordered
- Shift K into cells: albuterol or IV sodium bicarbonate as ordered
- Remove K: diuretic, sodium polystyrene sulfonate, or dialysis

WHY / DANGER K changes excitability -> rhythm + muscle problems. HYPERK: lethal dysrhythmia is the immediate danger.

DIET LOW: encourage potato, spinach, avocado, beans, tomato sauce, banana, orange juice. HIGH: limit; avoid salt substitutes.

BETTER? K trends to 3.5-5.0; rhythm stabilizes; muscle strength, respirations, and bowel sounds normalize.